

Mohamad Sanas Mohroof

Computer Engineering Undergraduate — AI & Machine Learning

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PROFILE

Computer Engineering undergraduate at the University of Jaffna with hands-on experience designing and deploying AI, machine learning, computer vision, and Retrieval-Augmented Generation (RAG) systems across 4+ end-to-end projects. Skilled across the full ML lifecycle: data preprocessing, feature engineering, model training and evaluation, explainability (SHAP), REST API deployment, and production AI application development using FastAPI, Flask, and LangChain. Seeking an internship in AI Engineering, Machine Learning Engineering, or Agentic AI / LLM Applications.

EDUCATION

University of Jaffna

Expected Graduation: April 2027

Bachelor of Science in Engineering (Hons.) in Computer Engineering

EXPERIENCE & LEADERSHIP

Vice President

Feb 2026 – Present

Computer Engineering Society, University of Jaffna

Kilinochchi, Sri Lanka

- Lead a team of [X] committee members in planning and coordinating [X]+ technical sessions, workshops, and student engagement events per semester.
- Partner with the executive committee to design skill-building initiatives, increasing student participation in technical activities by [X]%.

RESEARCH

Multi-Modal Social Media Analytics Framework for Colombo, Sri Lanka

2026 – Present

Undergraduate Research Project

- Designing a multimodal analytics framework to quantify urban public perception in Colombo from [X]+ social media posts combining text and image data.
- Integrating NLP, computer vision, image captioning, multimodal fusion, and spatial sentiment analysis to correlate public sentiment with geographic location.
- Applying SHAP-based explainability and correlation analysis to improve model interpretability and validate feature importance.
- **Technologies:** Python, PyTorch, TensorFlow, Transformers, OpenCV, SHAP, Machine Learning, Deep Learning.

PROJECTS

🔗 StudyGen AI

FastAPI — Next.js — LangChain — ChromaDB — PostgreSQL — Gemini/OpenAI

- Engineered an AI-powered study assistant using Retrieval-Augmented Generation (RAG), processing [X]+ documents and reducing manual study-prep time by [X]%.
- Built a document processing pipeline handling PDF extraction, chunking, embedding generation, and semantic retrieval with [X]ms average query latency.
- Implemented AI-generated summaries, quizzes, flashcards, and document-grounded Q&A, achieving [X]% answer accuracy on test queries.
- Architected a modular FastAPI backend with ChromaDB vector storage and a Next.js frontend, supporting [X] concurrent users.

🔗 ML-Based Credit Scoring System

Python — Scikit-learn — Flask — SHAP — Flutter Web

- Built an end-to-end machine learning pipeline for loan eligibility prediction, achieving [X]% model accuracy on a dataset of [X]+ records.
- Developed and deployed RESTful APIs with Flask, serving real-time predictions with [X]ms average response time.
- Integrated SHAP explainability to surface top predictive features, improving prediction transparency for [X] stakeholder(s).
- Deployed the full-stack application to production on Render, reducing manual credit-review time by [X]%.

🔗 Medical Report & Prescription Cross-Checker

FastAPI — LangChain — ChromaDB — React

- Built a RAG system enabling natural-language querying across [X]+ medical reports.
- Implemented multi-document semantic search using vector embeddings, improving retrieval relevance over keyword search.
- Developed confidence-scored medical question answering to flag low-certainty responses for human review.
- Designed a modular, provider-agnostic architecture supporting multiple LLM backends (OpenAI, Gemini) for cost and reliability flexibility.

🔗 Smart Supermarket Product Identification System

Python — OpenCV — NumPy — Matplotlib

- Built an image-processing pipeline to automate product identification, achieving [X]% classification accuracy across [X] product categories.
- Implemented preprocessing, segmentation, and classification stages, reducing manual inventory counting time by [X]%.
- Automated category-wise product counting with statistical visualizations for inventory reporting.
- Produced annotated outputs with bounding boxes and summary reports for [X]+ test images.

TECHNICAL SKILLS

Programming	Python, Java, C++, PHP, SQL, Dart
Machine Learning	Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation, SHAP, Classification, Regression
Generative AI & LLMs	LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, ChromaDB, Gemini API, OpenAI API
Computer Vision	OpenCV, Image Processing, Image Segmentation, Image Classification
Backend Development	FastAPI, Flask, Spring Boot, REST APIs
Databases	PostgreSQL, SQLite, ChromaDB
Tools & Platforms	Git, GitHub, Postman, Jupyter Notebook, VS Code

RELEVANT COURSEWORK

- Artificial Intelligence
- Machine Learning
- Probability and Statistics
- Data Structures and Algorithms
- Database Systems
- Software Engineering

CERTIFICATIONS

- Introduction to Artificial Intelligence
- Introduction to Software Engineering
- Python for Data Science and AI Development
- IBM – Developing AI Applications with Python and Flask
- Develop Generative AI Applications
- DeepLearning.AI – Supervised Machine Learning: Regression and Classification

SOFT SKILLS

Teamwork • Leadership • Problem Solving • Critical Thinking • Communication • Time Management • Adaptability

LANGUAGES

English	Professional Working Proficiency
Tamil	Native
Sinhala	Limited Working Proficiency

REFERENCES

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